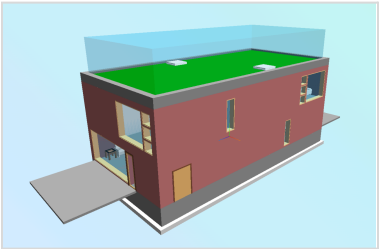
Take-Home Assignment

PhD Position: AI-based Generation of Building Designs

Task: Semantic and Geometric Information Embedding from an IFC Model



Objective: You are provided with an IFC file representing a building. Develop a conceptual framework for preparing and representing the semantic and geometric information contained in this model so that it can support AI methods such as graph neural networks, embedding models, geometric deep learning, neuro-symbolic systems, or other AI methods. The proposed framework should be suitable for AI-assisted design workflows, such as co-pilot systems for buildings.

Dataset: https://github.com/DiellzaElshani/ExampleData (provided for illustration purposes; the proposed framework should be applicable to other IFC files).

Important: This is not an implementation exercise. We are interested in your ability to think abstractly about data representation. Please prioritize clarity and depth of reasoning over breadth.

Aspects to Consider:

- **Semantic Representation:** Which types of semantic information are most meaningful for learning, and how should they be structured?
- **Geometric Representation:** How can geometric information be encoded so that it is both informative and computationally manageable?
- **Intended AI Capability:** What downstream AI capability should your representation enable, and why is your design suitable for it?
- **Data Transformation:** How would you transform and prepare the information contained in the IFC to support your proposed representation and ensure it is usable and consistent for the chosen AI method?
- **Symbolic Constraints:** How can symbolic constraints be incorporated into the representation or learning pipeline?
- **Additional Considerations:** Include any additional considerations you consider important.

Deliverable: *Submit a PDF document of no more than two pages, with diagrams strongly encouraged to illustrate your concepts. Code and external repositories must not be included. Please include relevant scientific references where appropriate.*

Evaluation Criteria: *Evaluation will focus on abstraction ability, system-level reasoning, representation coherence, understanding of AI-ready data, and technical justification.*

Submission Deadline: *You have up to 48 hours from the time you receive this email to complete the assignment. Please submit your document to diellza.elshani@icd.uni-stuttgart.de and thomas.wortmann@icd.uni-stuttgart.de.*